

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 1

Paper 3: Computation Emerges: Surround-Driven Repair, Coherence Bistability, and the First Subsystem

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Paper 3

Computation Emerges: Surround-Driven Repair, Coherence Bistability, and the First Subsystem

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Status of Results

This paper distinguishes four categories of claim, each labeled where it appears in the text.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and the structures established in Papers 1 and 2 by explicit construction or proof. Surround-driven repair as a direct consequence of FCC boundary geometry, requiring no extension of the $[[4,2,2]]$ repair operator (§ 2.2); the Imprinting Theorem stating that a region of diameter L_R converges to the unique stabilizer-satisfying continuation of its retarded surround history over imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$ flops (§ 2.3); Hebbian co-firing as a corollary of the imprint mechanism under structured boundary conditions (§ 2.4); the dissolution rate of non-T-closed configurations and the effective error-rate decay $(1/3)^d$ (§ 2.5, closing the Paper 2 deferrals); the recursive applicability of the axioms at the coarse-grained scale L_p , producing the meta-level triadic rule, meta-level FCC structure, and meta-level $[[4,2,2]]$ stabilizer dynamics by the same arguments applied at the same axiomatic content (§ IV); the (C_D, C_O, C_C) trichotomy as the unique minimum sufficient decomposition of any non-trivial information-transforming configuration (§ 5.2); the minimum subsystem extent $L_{\text{sub}} = 3L_p + \delta$ (§ 5.3); the existence of a minimum computing subsystem by explicit construction (§ 5.4); and the information-generation rate differential of one novel P-state per meta-flop per minimum subsystem (§ 5.5) are derived in this sense.

Structural correspondence. The coherence bistability threshold $|\Pi_\psi|_*(L_R) = 1 - 2\sqrt{c_{\text{TMS}}/(6 L_R)}$ of § 3.3 is a derivation of the scaling form, not of the exact coefficient. The functional dependence $p_{\text{error}} \propto 1/\sqrt{L_R}$ is forced by balancing per-flop noise injection against per-flop imprinting capacity under FCC connectivity. The numerical coefficients depend on combinatorial factors (notably the $C(4,2) = 6$ upper bound on per-tetrahedral-unit two-error configurations) and geometric proportionality factors on the imprinting side, which are stated as worst-case bounds and reserved for refinement in Paper 5’s simulation work. The threshold’s qualitative behavior — strictly between 0 and 1, monotonic in L_R , producing a bistable phase structure — is forced by the derivation. Its quantitative pinning is not.

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose closed forms are not derived within this paper. The hierarchical lightspeed formula $c^{(k)}$ (§ 6.1) gives the structural form, with the $L_p(k)/L_p(k-1)$ and $T_p(k)/T_p(k-1)$ ratios identified as combinatorial quantities requiring enumeration of stabilizer-satisfying T-closed configurations at each level. The refined $\beta = f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}})$ prediction (§ 6.2) specifies the functional dependences; f ’s explicit form is deferred. The spatial-

distribution prediction for computational activity in old substrate regions (§ 6.3) is a structural claim whose empirical bridge requires the simulation work of Paper 5. The structural parallel to self-organized criticality flagged in § 3.5 is a positioning claim; specific statistical predictions (power-law cascade-collapse statistics, scale-free correlation lengths, $1/f$ signatures in novel-configuration emergence rates) are conjectures pending Paper 5's simulation work.

Deferred derivations carried into Papers 4 and 5. The minimum program extent L_p , characterized in § 4.1 as approximately 4–8 lattice spacings, is left as a placeholder pending combinatorial enumeration of stabilizer-satisfying T-closed configurations at substrate level. The exact integer value, and the corresponding $L_p^{(k)}$ at higher hierarchical levels, are reserved for Paper 5. The closed form of f and the explicit numerical $c^{(k)}$ sequence inherit this deferral. The macroscopic consequences of subsystem dynamics — hierarchies, multi-subsystem dynamics, novelty exhaustion, the dispersion structure of $c^{(k)}$ — are carried into Paper 4. The cross-level relativistic phenomena flagged in § 6.1 are carried into Paper 4 § 8.2 and resolved structurally in Paper 5.

This four-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category.

Preface to Paper 3

This paper is the third installment in Volume 1 of the Triadic Measurement Substrate program. Volume 1, Paper 1 established the geometric and ontological architecture of the substrate — the agent manifold $M\alpha$, triadic confirmation, FCC necessity, the $[[4,2,2]]$ structural correspondence, and the wave dynamics of the confirmation and polarization fields. Volume 1, Paper 2 developed the substrate's computational dynamics — the polarization-biased resolution of synthetic bits, the Born-rule structural correspondence, the 3-input majority gate, and the formal definition of programs as configurations causally closed under the flop operator T with $K(C) < |C|$. Paper 3 takes both as given and develops the next move: how computation emerges from informational substrate, how learning is forced by the geometry rather than added to it, and how the first subsystem — the smallest configuration that transforms patterns rather than merely persisting them — is generated by the substrate dynamics without external design.

As in Papers 1 and 2, no new axioms are introduced. No new free parameters are introduced. The Principle of Generative Economy continues to govern every selection. Paper 3 additionally closes several formal debts deferred from Paper 2 — the dissolution rate of non-T-closed configurations, the effective error-rate decay with coherence depth, and the existence and form of the coherence bistability threshold — by derivations that follow from the same geometric primitives that produced wave dynamics and biased measurement.

Abstract. Building on the binary alphabet $\bar{B} \in \{0, 1\}$, the polarization field ψ , and the program formalism established in Paper 2, we develop the computational dynamics of the TMS substrate at and above the program scale. Five principal results are derived. First, the $[[4,2,2]]$ stabilizer repair operator of Paper 1, when applied at boundary sites of any finite region R , automatically incorporates the polarization landscape of R 's surround into the repair bias — by the FCC geometry of parent-child connectivity alone, with no new mechanism introduced. This produces the Imprinting Theorem: over the imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$ flops, R 's polarization landscape converges to the unique stabilizer-satisfying continuation of the retarded surround history. Hebbian co-firing follows as a corollary. Second, the dissolution rate of non-T-closed configurations and the effective error-rate decay with coherence depth — both deferred from Paper 2 — are derived from the same iteration as $O(L/\ell_p)$ flops and $(1/3)^d$ respectively. Third, the per-event error probability is shown to be a function of local polarization coherence, and balancing per-flop noise injection (combinatorially set by $[[4,2,2]]$ error statistics) against per-flop imprinting capacity (geometrically set by FCC connectivity) yields a closed-form coherence bistability threshold $|\Pi_\psi|_*(L_R)$ separating regions that sustain program-bearing algorithmic content from regions that decay toward noise. Fourth, the substrate's axioms are scale-independent: applied at the coarse-grained scale L_p of minimum stabilizer-satisfying T-closed configurations, they produce the same triadic rule, the same FCC geometry, and the same $[[4,2,2]]$ stabilizer structure at the meta-level — hierarchy by geometric necessity. Fifth, the minimum subsystem is derived as a composite of three programs (data, operation, control) at meta-tetrahedral extent $L_{\text{sub}} = 3 L_p + \delta$, with the trichotomy shown to be the unique minimum sufficient decomposition of any non-trivial information-transforming configuration. The substrate concentrates computation into bit-rich regions, generates subsystems wherever the local disruption rate favors information-generating configurations, and supports a recursive hierarchy with derived (generically distinct) lightspeeds at each level. No new free numerical parameters are introduced; every quantity follows from the axioms and the Principle of Generative Economy.

I. Inheritance from Papers 1 and 2

1.1 Inherited Structure

Paper 3 takes the following structures from Papers 1 and 2 as given:

- The agent manifold $M\alpha$ with agent spacing $\ell_0 \equiv \ell_p$, and the FCC causal stack with face-diagonal nearest-neighbor connectivity (Paper 1 § 2–§ 3).
- The triadic actualization rule $\bar{B} = 3\bar{\alpha}_2$ in 2D and $\bar{B} = 4\bar{\alpha}_3$ in 3D, with the $1 \rightarrow 4$ multiplier and informational closure (Paper 1 § 2, § 4).
- The binary alphabet $\bar{B} \in \{0, 1\}$ with polarization $\chi \equiv 2\bar{B} - 1 \in \{-1, +1\}$, and polarization-carrying touch vector signals (Paper 1 § 1.3, § 1.4).
- The flop as the atomic unit of state change, with no sub-flop time (Paper 1 § 1.5).
- The wave equations for the confirmation density ϕ and the polarization field ψ at $c_{\text{TMS}} = 1/\sqrt{2}$ (Paper 1 § 6; Paper 2 § 2).
- The $[[4,2,2]]$ structural correspondence: logical qubit LQ1 = bit value, LQ2 = causal-history inheritance, jointly protected by stabilizers S_x and S_z at distance 2 (Paper 1 § 5).
- The polarization-biased resolution rule $P(\bar{B} = 1 | \Pi_\psi) = (1 + \Pi_\psi)/2$ and its structural correspondence to the Born rule (Paper 2 § 4).
- The 3-input majority gate as the substrate's irreducible computational primitive and the universality result $\{\text{MAJ, constant}\} \rightarrow \{\text{AND, OR, NOT}\}$ (Paper 2 § 5).
- The flop operator T and the formal definition of a program: a configuration C with $T^n[C] = C$ and $K(C) < |C|$ (Paper 2 § 6.2).

- U_{sh} as the pervasive maximum-entropy flux present at every depth of the causal stack, providing both the source of latent bits and the stochastic disruption to confirmed states (Paper 1 § 1.1, § 6.3).

1.2 The Computational Questions Paper 3 Answers

Paper 2 established that the substrate computes — the majority gate is the irreducible primitive, universality is forced by the gate set, and programs are configurations causally closed under T with $K(C) < |C|$. Paper 2 deferred three formal derivations to Paper 3: the dissolution rate of non-T-closed configurations, the effective error-rate decay with coherence depth, and the explicit conditions under which the substrate selects program-bearing configurations against U_{sh} disruption. Paper 3 answers these and develops three further structural questions: how the substrate learns from its environment, how computational density concentrates into bit-rich regions, and how the first subsystem — the smallest configuration that transforms patterns rather than merely persisting them — emerges from the same geometric primitives without additional design.

II. Learning as Imprinted Repair

2.1 The Surround

Definition 0.0.1 (Surround). For a finite region R of the FCC lattice, the surround S_R is the set of lattice sites within nearest-neighbor touch-vector range of the boundary ∂R that lie outside R itself:

$$S_R = \{y \notin R : \exists x \in \partial R \text{ with } y \in \text{neigh}(x)\}$$

The polarization landscape on S_R is the time-dependent function $\psi_S(\tau) = \{\chi(y, \tau) : y \in S_R\}$. The touch vector network at ∂R carries signals from S_R into R at every flop. By the connectivity established in Paper 1 § 1.3, no other channel of informational communication between R and the rest of the lattice exists. The surround is the complete informational environment of R .

2.2 Surround-Driven Repair

Paper 1 § 5.2 established that when U_{sh} disrupts a confirmed bit in R , the synthetic bits produced at the next flop detect the stabilizer syndrome and repair the missing or anomalous confirmation by projecting toward the nearest stabilizer-satisfying state. Paper 2 § 5.2 treated “nearest” as a local concept — the nearest configuration consistent with R ’s own interior. That treatment was sufficient at the level of universality but incomplete at the level of repair geometry.

Proposition 0.0.2 (Surround Dependence of Repair). *The stabilizer-repair projection at a boundary site $x \in \partial R$ has a parent set that includes at least one agent in S_R . The polarization landscape on S_R therefore contributes to the bias of the repair; and the “nearest stabilizer-satisfying configuration” is determined jointly by the polarization landscapes on R and S_R .*

Proof. By Paper 1 § 4.2, a synthetic bit at position x has three parent agents in the prior layer, and its polarization is biased by the three incoming touch vector signals from those parents (Paper 2 § 3.2). For $x \in \partial R$, the FCC geometry forces at least one of x ’s parents to lie in S_R rather than in R — this is the geometric definition of a boundary site. The polarization at that parent contributes to $\Pi_\psi(x)$ by the same linear weighting that governs all parent-to-child polarization transfer. The repair projection at x is therefore biased by the polarization on S_R through this parent.

□

Corollary 0.0.3 (No New Mechanism). *Surround-driven repair requires no extension of the repair operator beyond what Paper 1 § 5 already established. The same mechanism — synthetic bits at the next flop carrying their parent polarizations into the resolution of a syndrome violation — automatically includes surround contributions at boundary sites by FCC geometry alone. What Paper 2 treated as local was already non-local by construction.*

This is the key recognition. The $[[4,2,2]]$ repair is a projection operator whose bias is set by the full set of parent polarizations at the repair site. At interior sites of R , all parents lie in R and the repair is local. At boundary sites, parents lie in both R and S_R and the repair is surround-driven. The repair operator is the same; its locality is a property of the site, not of the mechanism.

2.3 Imprinting Over the Retarded Surround History

The boundary sites of R communicate surround information into R 's interior at each flop. Iterating the repair operator over successive flops propagates this information inward at the lattice propagation speed $c_{TMS} = 1/\sqrt{2}$ (Paper 1 § 6.5). For a region of diameter L_R , complete inward propagation requires $d_R = \sqrt{2} \cdot L_R/\ell_p$ flops. We refer to d_R as the imprint depth of R .

The polarization information arriving at any interior site of R at flop τ is not the value of ψ_S at that moment, but the value of ψ_S at retarded time $\tau - d/c_{TMS}$, where d is the lattice distance from the relevant boundary point to the interior site. This is forced by signal conservation (Axiom 3) and the finite propagation speed established in Paper 1 — it is not an assumption added here. The imprint at flop τ is therefore a function of the retarded surround history over the window $[\tau - d_R, \tau]$.

Definition 0.0.4 (Imprinted Configuration). The configuration of R after $k \geq d_R$ flops of repair under the surround history $\psi_S(\tau')$, $\tau' \in [\tau - d_R, \tau]$, is the imprinted configuration $\psi_R * (\psi_S|[\tau - d_R, \tau])$ — the unique stabilizer-satisfying continuation of that retarded history into R .

Theorem 0.0.5 (Imprinting Theorem). *The imprinted configuration $\psi_R * (\psi_S|[\tau - d_R, \tau])$ is the unique polarization landscape on R that minimizes the expected per-flop stabilizer syndrome rate consistent with the retarded surround history at the boundary.*

Proof sketch. The stabilizer repair operator is a contraction on the space of polarization landscapes: configurations farther from stabilizer parity are projected closer to it at each flop. The fixed points of the repair operator are exactly the stabilizer-satisfying configurations. With the retarded boundary history fixed, the iterated repair operator converges to the stabilizer-satisfying configuration whose polarization landscape is consistent with the retarded surround history at the boundary and stabilizer-satisfying in the interior. This configuration is unique because the $[[4,2,2]]$ parity conditions in a coherent region determine the polarization at each interior site given its retarded boundary values up to discrete flips, and the boundary values are fixed by the retarded ψ_S . $\square$

The configuration of R after $k \geq d_R$ flops is therefore not a function of R 's initial state but a function of the surround's history. R 's polarization landscape encodes the surround's history at the boundary, automatically averaged across propagation delays. When ψ_S varies slowly compared to d_R , R tracks ψ_S closely. When ψ_S varies rapidly, R averages over the variation. No regime selector is needed — the retarded average is what the geometry produces. This encoding is what learning is.

2.4 The Hebbian Corollary

Hebbian co-firing — the principle that simultaneous activity in connected units strengthens their effective bond (Hebb, 1949) — is the canonical mechanism by which environmental correlations are encoded in a neural substrate. It is typically presented as a phenomenological rule for synaptic plasticity. Recent work in statistical mechanics has derived Hebbian learning from a maximum-entropy variational principle (Albanese et al., 2024), establishing it as a theorem about how networks extract correlations under entropy extremization. In the TMS, Hebbian co-firing emerges as a different and complementary theorem — about how the geometric stabilizer-repair operator behaves under structured boundary conditions. Both derivations target the same phenomenon; the convergence of independent first-principles derivations strengthens the result.

Corollary 0.0.6 (Hebbian Co-Firing). *Let $s_i, s_j \in S_R$ be two surround sites and let $r \in R$ be an interior site causally downstream from both. Define the surround correlation over k flops:*

$$Cij(k) = (1/k) \sum \tau = 1^k \chi(s_i, \tau) \cdot \chi(s_j, \tau)$$

and the bond strength between (s_i, s_j) and r :

$$Wij,r(k) = (1/k) \sum \tau = 1^k \chi(r, \tau) \cdot (\chi(s_i, \tau) + \chi(s_j, \tau))/2$$

Then $Wij,r(k)$ is monotonically increasing in $|Cij(k)|$ for $k \geq d_R$.

Proof. By Paper 2 § 3.2, the polarization at r is determined by its parents via the linear bias rule. The parents of r at flop τ have causal histories tracing back to S_R at flop $\tau - d$, where d is the causal depth from S_R to r . When s_i and s_j co-fire positively (both $+1$ or both -1), the polarization signals propagating from them through the lattice reinforce each other at every intermediate layer; their joint contribution to Π_ψ at r 's parents is amplified relative to the case of uncorrelated firing. When they anti-fire, their contributions partially cancel. The bond strength $Wij,r(k)$ uses the linear average $(\chi(s_i) + \chi(s_j))/2$ — the same linear weighting that the substrate uses at every parent-to-child transfer — and is therefore the natural measure consistent with the substrate's own dynamics, with no additional non-linear operation introduced. Iterated over k flops with $k \geq d_R$, the time-averaged bias at r tracks the time-integrated correlation $Cij(k)$. The stabilizer repair operator, by § 2.2, projects r toward configurations consistent with this bias. The bond strength $Wij,r(k)$ therefore increases monotonically with $|Cij(k)|$. □

Remark 0.0.7. The Hebbian rule in standard neural network theory is a designed update — a learning rule chosen to produce desired representational properties. In the TMS, the Hebbian rule is a theorem about how the geometric repair operator behaves under structured boundary conditions. The substrate learns not because it has been equipped with a learning mechanism, but because the repair mechanism it was already equipped with for fault tolerance has surround-imprinting as its non-local extension. Learning and fault tolerance are the same geometric process applied at different spatial scales.

2.5 The Convergence Rate and the Dissolution Rate

The imprinting process converges over the imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$ flops. This is the same timescale that governs the dissolution of non-closed configurations promised at Paper 2 § 6.3. We derive both here from the same argument.

Theorem 0.0.8 (Dissolution and Convergence). *Let C be a configuration on R of spatial extent L_R .*

- If C satisfies stabilizer parity and is causally closed under T (C is a program in the sense of Paper 2 § 6.2), then C persists indefinitely against single-bit disruptions from U_{sh} .
- If C satisfies stabilizer parity but is not causally closed under T, then C dissolves into the imprinted configuration $\psi_R * (\psi_S[\tau - d_R, \tau])$ over the imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$ flops.
- If C does not satisfy stabilizer parity, then C is repaired toward the nearest parity-satisfying configuration within a single flop, then evolves according to (i) or (ii).

Proof sketch. Case (i) is Paper 1 § 5.3. Case (iii) is Paper 1 § 5.2. For case (ii), C is a fixed point of the repair operator at flop τ but not of the flop operator T. The next-flop state $T[C]$ differs from C; the difference propagates outward at c_{TMS} from the sites of disagreement. Over k flops, the cumulative drift is $O(k \cdot c_{TMS} \cdot \ell_p)$. When this drift exceeds L_R , the configuration has been entirely replaced by whatever the retarded surround history supplies. That threshold is reached at $k = L_R / (c_{TMS} \cdot \ell_p) = \sqrt{2} \cdot L_R / \ell_p$. The resulting configuration is $\psi_R * (\psi_S[\tau - d_R, \tau])$. $\square$

Corollary 0.0.9 (Effective Error-Rate Decay). *In a coherent region where ψ_S is structured and stationary over d_R flops, the probability that a single-bit error survives stabilizer repair over d successive flops is bounded above by:*

$$P_{\text{error}}(d) \leq (1/3)^d$$

with the bound becoming tight when the coherence region is large compared to the error correlation length (one lattice spacing). For smaller coherence regions, repair rounds at successive layers operate on overlapping tetrahedral units and the bound is loose but still upper.

Proof. The per-event error probability in the intermediate $|\Pi_\psi| = 1/3$ case is $1/3$ (Paper 2 § 5.1). Repair rounds at successive layers are independent in the limit of large coherence regions because the repair at layer $\tau + 1$ operates on synthetic bits whose parents differ from those at layer τ . The probability of d independent uncorrected errors is the d -th power of the per-event probability. $\square$

These results settle both formal debts carried forward from Paper 2. The dissolution rate and the error-decay rate are not independent results — they are two readings of the same geometric process. Configurations dissolve into surround-imprinted forms over $O(L/\ell_p)$ flops; single-bit errors are corrected at the same rate. Learning and fault tolerance are the same operation at the same speed.

III. Coherence Phases and the Concentration of Computational Density

3.1 Bit-Richness as Polarization Structure

Throughout this paper we use the term bit-rich to refer specifically to regions whose polarization landscape sustains programs in the sense of Paper 2 § 6.2 — configurations causally closed under T with $K(C) < |C|$. Bit-richness is a property of the polarization landscape across confirmed sites, not of the confirmation density ϕ . A region with ϕ near 1 but uniform polarization carries no programs; a region with moderate ϕ but coherent polarization landscape may carry many. Confirmation density and polarization structure are independent axes (Paper 2 § 2.4), and bit-richness lives on the second.

The three-state distinction established in Paper 1 § 1.4 — latent (in U_{sh} , undecided), confirmed-zero ($\bar{B} = 0$), and confirmed-one ($\bar{B} = 1$) — is the substrate’s irreducible computational state space. φ measures confirmed-vs-latent (either value of $\bar{B}$ counts toward φ). Polarization structure measures the pattern of 0s and 1s among already-locked bits. “Bit-rich” as used in this paper refers exclusively to the second.

The local algorithmic content of a region is bounded above by its confirmation density:

$$K_{\psi}(R) \leq N_R \cdot \log 2 = \varphi_R \cdot V_R \cdot \log 2$$

This counting bound holds for any configuration. Hosting programs of substantial algorithmic content requires both confirmed substrate (high φ) and structured polarization landscape across that substrate. Programs sit strictly between the two extremes of trivially uniform polarization ($K_{\psi} \approx \log V_R$) and algorithmically incompressible polarization at the local U_{sh} rate ($K_{\psi} \approx |\text{Cl}| \cdot \log 2$).

3.2 The Per-Event Error Rate as a Function of Coherence

The per-event error rate introduced in Paper 2 § 5.1 is not a fixed constant. It is a function of the local polarization amplitude. From the resolution probability $P(\bar{B} = 1 \mid \Pi_{\psi}) = (1 + \Pi_{\psi})/2$, the probability that a confirmation resolves contrary to the dominant polarization of its parents is:

$$p_{\text{error}}(\Pi_{\psi}) = (1 - |\Pi_{\psi}|)/2$$

For $|\Pi_{\psi}| = 1$ (fully aligned parents), $p_{\text{error}} = 0$. For $|\Pi_{\psi}| = 0$ (maximally disagreed parents), $p_{\text{error}} = 1/2$. For $|\Pi_{\psi}| = 1/3$ (two-vs-one parents), $p_{\text{error}} = 1/3$, recovering the value used in Paper 2 § 5.1.

The error rate is therefore set by local polarization coherence — the degree to which parent polarizations at each confirmation event agree. Coherent regions have low p_{error} ; incoherent regions have high p_{error} . There is no separately specified disruption mechanism. What Paper 1 § 6.3 referred to as “ U_{sh} regulation” is precisely the residual stochasticity in the polarization-biased resolution, and its local magnitude is governed by the polarization landscape itself.

A region’s per-flop accumulation of incompressible noise scales as $N_R \cdot p_{\text{error}}(|\Pi_{\psi}|)$ — the count of confirmed sites times the local error probability. When p_{error} is small, the noise injection is small relative to the $[[4,2,2]]$ repair capacity, and the algorithmic structure of ψ is preserved. When p_{error} is large, noise injection exceeds repair capacity, and structured polarization patterns decay toward incompressibility.

3.3 The Critical Coherence Threshold

Theorem 0.0.10 (Coherence Bistability, Scaling Form). *A region R sustains program-bearing algorithmic content in its polarization landscape if and only if its local coherence $|\Pi_{\psi}|_R$ exceeds a critical threshold $|\Pi_{\psi}|_*(L_R)$ whose scaling with the region’s characteristic length is*

$$|\Pi_{\psi}|(L_R) \sim 1 - 2 \sqrt{c} \cdot \alpha_{\text{geom}} / (\beta_{\text{comb}} \cdot L_R))^*$$

where β_{comb} captures per-tetrahedral-unit error combinatorics (worst-case upper bound $\beta_{\text{comb}} = C(4,2) = 6$) and α_{geom} captures imprinting-side geometric proportionality (worst-case lower bound $\alpha_{\text{geom}} = 1$). In these worst-case bounds the threshold takes the closed form

$$|\Pi_{\psi}|(L_R) = 1 - 2 \sqrt{c_{\text{TMS}}} / (6 L_R))^*$$

which is strictly between 0 and 1 for all $L_R \geq 4\ell_p/(6 c_{\text{TMS}})$.

Proof. Each tetrahedral confirmation unit contains 4 agents, and the $[[4,2,2]]$ code has distance 2: single-agent errors per unit are correctable, two-or-more-agent errors per unit are not. With per-event error probability $p_{\text{error}} = (1 - |\Pi_\psi|)/2$, the per-unit uncorrectable error probability is approximately $P_{\text{unc}} \approx 6 p_{\text{error}}^2$ for moderate p_{error} , with the factor 6 = C(4,2) counting the ways two errors can occur among four agents in a tetrahedral unit. Each uncorrectable error injects incompressibility into ψ at that site at a rate that scales as $V_R \cdot P_{\text{unc}}$ per flop, where V_R is the region's volume in lattice units.

The imprinting mechanism (§ II) counteracts this loss. The retarded surround history is communicated into R through its boundary at rate $A_R \cdot c_{\text{TMS}}$, where A_R is the boundary area in lattice units. The stabilizer-satisfying continuation of the retarded surround reduces local entropy at a rate proportional to this flow.

Balancing per-flop noise injection against per-flop imprinting gain yields the self-consistent equation:

$$6p_{\text{error}}^2 \cdot V_R = A_R \cdot c_{\text{TMS}}$$

For a region of characteristic length L_R , $A_R/V_R \sim 1/L_R$, giving $p_{\text{error}}^2 = c_{\text{TMS}}/(6 L_R)$, and therefore:

$$p_{\text{error, crit}} = \sqrt{c_{\text{TMS}}/(6 L_R)}$$

Translating back to the coherence: $|\Pi_\psi|(L_R) = 1 - 2 p_{\text{error, crit}} = 1 - 2 \sqrt{c_{\text{TMS}}/(6 L_R)}$. The threshold is strictly between 0 and 1 for all finite L_R because $c_{\text{TMS}}/(6 L_R)$ is positive and bounded above by 1/4 in the regime of interest $L_R \geq 4\ell_p/(6 c_{\text{TMS}})$. Small regions have proportionally more surround influence and lower thresholds; large regions have less and higher thresholds.

□

The threshold is geometric: it follows from the $[[4,2,2]]$ code distance, the polarization-biased resolution rule, and the FCC connectivity structure. The functional dependence on L_R is forced by the derivation; the numerical coefficients inside the square root are stated as worst-case bounds.

Remark 0.0.11 (Status of the Threshold). The derivation above forces the scaling form $p_{\text{error}} \propto 1/\sqrt{L_R}$ by balancing per-flop noise injection (scaling as $p_{\text{error}}^2 \cdot V_R$) against per-flop imprinting gain (scaling as $A_R \cdot c$), under FCC connectivity and $[[4,2,2]]$ code distance 2. The functional dependence on L_R is therefore derived. The numerical coefficients inside the square root are not — they depend on combinatorial and geometric factors stated here as worst-case bounds. The factor $\beta_{\text{comb}} = 6$ is the C(4,2) count of two-error configurations in a tetrahedral unit; the true effective value is generically lower under correlated error statistics. The factor $\alpha_{\text{geom}} = 1$ absorbs the imprinting-side geometry; the true effective value depends on the precise boundary topology of R and may carry a multiplicative correction of order unity. Both refinements are reserved for Paper 5's simulation work.

The structural content of the theorem — that a critical threshold exists, that it scales as $1 - 2\sqrt{\cdot}/L_R$, that it is strictly between 0 and 1, that small regions face higher thresholds than large ones, that the threshold separates a bistable phase structure — is forced by the derivation. The exact numerical pinning of the threshold for a given L_R is not. Section 3.4's self-reinforcement result and § 3.5's continuous-sampling framing depend on the structural content; they do not depend on the precise coefficient values.

3.4 Self-Reinforcement of Coherent Regions

Theorem 0.0.12 (Self-Reinforcement of Coherent Regions). *Let R be a region with local coherence $|\Pi_\psi|_R > |\Pi_\psi|(L_R)$ and let S_R be its surround. Over $k \geq d_R$ flops, the imprinting mechanism of § II drives R*

toward the configuration $\psi_R(\psi_S[\tau - d_R, \tau])$ that minimizes the per-flop syndrome rate. The resulting configuration has coherence $|\Pi_\psi|/R^* \geq |\Pi_\psi|_R$. *Coherent regions become more coherent; incoherent regions remain incoherent and dissolve toward noise.*

Proof. By the Imprinting Theorem (§ 2.3), R relaxes toward the stabilizer-satisfying continuation of the retarded surround history. Stabilizer-satisfying configurations have by definition the lowest possible local syndrome rate. By § 3.2, low syndrome rate corresponds to high local coherence $|\Pi_\psi|$. The imprinted configuration is therefore at least as coherent as R’s initial state, and strictly more coherent unless R was already at a fixed point of imprinting under its current surround. A region above the threshold therefore becomes more coherent over flop time and remains above the threshold; a region below the threshold cannot stabilize, as its low coherence produces an uncorrected error rate exceeding the imprinting capacity, and its program content decays toward U_{sh} — like noise. The two phases are dynamically separated. □

The TMS does not produce a uniform sea of computation. It produces sharp regions of coherent, program-bearing structure separated by incoherent intervening medium. The boundaries between these phases are themselves geometric features of the substrate, governed by the propagation of polarization waves across the critical-coherence interface.

3.5 The Substrate as Continuous Sampler

Sections 3.3 and 3.4 together establish a strong claim about the substrate’s long-time behavior. U_{sh} is the pervasive maximum-entropy flux present at every depth of the causal stack (Paper 1 § 1.1, § 6.3). By Axiom 0, U_{sh} is algorithmically incompressible, with no preferred direction, scale, or target. The substrate is therefore continuously sampling all possible polarization configurations — every possible pattern is being attempted, everywhere, all the time. The dynamics select among these attempts.

Configurations above the coherence threshold self-reinforce by the imprinting plus bistability mechanisms of § 3.3 and § 3.4. Configurations below the threshold decay back into U_{sh} — like noise. The substrate does not produce coherent regions through any special generative mechanism; it retains them while letting everything else dissolve. The selection is the mechanism. U_{sh} is the source.

This framing is consistent with the pervasive flux character of U_{sh} established in Paper 1: U_{sh} is not only at the leading edge of the cascade but at every depth, which is why the $[[4,2,2]]$ repair must fire continuously to preserve already-confirmed bits against ambient disruption. Coherent regions form wherever the substrate’s own dynamics produce a configuration that the substrate’s own repair operator can sustain.

Computation and confirmation are not two separate processes. They are the same process under structured polarization conditions. Bit-rich regions are exactly the regions where confirmation can be sustained against U_{sh} — coherent parents reliably bias their children’s resolution, keeping p_{error} low, keeping uncorrectable errors below the imprinting rate, keeping the region intact. Bit-poor regions cannot sustain confirmation indefinitely; their high p_{error} means the resolution does not lock reliably, and the region returns to U_{sh} . The substrate computes by confirming, and confirms by computing. Anywhere it is not computing, it is not confirming; anywhere it is not confirming, it does not exist as substrate.

Remark 0.0.13 (Structural Parallel to Self-Organized Criticality). The dynamics described in § 3.3–§ 3.5 — a structurally supercritical amplification mechanism (the $1 \rightarrow 4$ cascade) balanced against a structurally subcritical disruption mechanism (U_{sh} flux at every depth), with the observable substrate sitting on the dynamic-stability boundary between them — rhymes with the architecture of self-organized criticality (Bak,

Tang, & Wiesenfeld, 1987; Jensen, 1998). SOC describes systems that naturally settle at the boundary between supercritical and subcritical regimes without external tuning, producing characteristic statistical signatures (power-law event distributions, scale-free correlation lengths, $1/f$ noise) as the empirical fingerprint of the critical state.

The TMS substrate exhibits the structural preconditions for SOC behavior: the cascade dynamics are geometrically forced to amplify, the U_{sh} disruption is geometrically forced to oppose, and the $[[4,2,2]]$ repair operator provides the local relaxation mechanism. The coherence bistability of § 3.3 is the substrate's expression of the critical state — regions above $|\Pi_\psi|_*(L_R)$ sustain structure, regions below dissolve, and the boundary between them is where the cascade and disruption balance dynamically. The continuous-sampling mechanism of § 3.5 is the SOC drive: U_{sh} continuously injects new configurations across the substrate; the geometric selection retains those above threshold and dissolves the rest.

One structural disanalogy is worth noting. Classical SOC depends on a temporal-scale separation between slow drive (e.g., sand grains added slowly to the Bak-Tang-Wiesenfeld pile) and fast relaxation (avalanches discharging accumulated stress quickly). TMS has continuous drive and continuous relaxation operating at the same atomic scale (every flop, every site). Whether this places TMS in the classical SOC universality class or in a non-classical SOC variant — and whether an emergent slow/fast separation arises at the level of bit-rich regions versus their boundaries — is an open question whose resolution depends on the cascade-collapse statistics derived in Paper 5's simulation work.

The structural parallel is flagged here as a research direction. Specific predictions — power-law statistics for cascade collapse events, scale-free correlation lengths in bit-rich region geometry, $1/f$ signatures in novel-configuration emergence rates — are testable in principle and will be developed alongside the combinatorial enumeration of stabilizer-satisfying T-closed configurations in Paper 5.

IV. Coarse-Graining and the Second-Order Alphabet

4.1 The Minimum Program Scale

By § 3.3, regions sustaining program-bearing polarization landscapes have a coherence threshold $|\Pi_\psi|_*(L_R)$ that depends on region size. The relation derived above places a lower bound on the spatial extent of any region capable of hosting a program against U_{sh} disruption: as L_R decreases, the required coherence increases, until the threshold approaches 1 and no configuration short of perfect alignment can sustain itself. The minimum program extent L_p is the smallest L_R for which a non-trivial T-closed configuration of compressibility $K(C) < |C|$ can stably exist.

Derivation of L_p . A non-trivial program requires at least one boundary between polarization domains — a configuration with $K(C) > \log V_R$ must contain at least one non-uniformity. The $[[4,2,2]]$ repair sustains each domain only if its thickness is large enough that boundary errors do not propagate inward faster than they can be corrected. The minimum thickness is set by the FCC nearest-neighbor distance (one lattice spacing) plus the propagation distance during one error-correction cycle (one further lattice spacing along a face diagonal). The minimum non-trivial program therefore has at least one domain boundary plus two stabilizing layers on each side, giving a minimum spatial extent of approximately 4 to 8 lattice spacings depending on geometry.

$$L_p \approx (4 - 8)\ell_p$$

The exact integer value of L_p for a given configuration shape is determined by closed-form enumeration of stabilizer-satisfying T-closed configurations under the $[[4,2,2]]$ code in the FCC lattice — a combinatorial

calculation tractable in principle and most efficiently approached through simulation alongside the program zoo enumeration deferred to Paper 5. The structural fact required here is that L_p is finite and strictly greater than ℓ_p : a single confirmed bit cannot be a program because $K(C) \approx |C|$ for any single-bit configuration; the minimum non-trivial program is several lattice spacings.

L_p plays the same role at the meta-level that ℓ_p plays at the substrate level. It is the minimum resolvable length of the coarse-grained system. Below L_p , programs do not exist; the substrate is structureless at that scale by the same architectural necessity that makes the substrate structureless below ℓ_p .

4.2 The Coarse-Graining Operator G

Definition 0.0.14 (Coarse-Graining Operator). Let R be a spatial region of the FCC lattice with diameter $L_R \geq L_p$, and let $[\tau_0, \tau_0 + n]$ be a time window covering one period of a candidate program. The coarse-graining operator G maps the polarization landscape ψ on $R \times [\tau_0, \tau_0 + n]$ to a single meta-level state $\bar{P}_p \in \{0, 1\}$ indexed by program identity p :

$$*G : \psi(R \times [\tau_0, \tau_0 + n]) \mapsto \bar{P}_p(X, T)^*$$

where X is the meta-spatial center of R (the centroid of R 's spatial support) and T is the meta-time index counting completed periods.

G evaluates to $\bar{P}_p = 1$ if ψ on $R \times [\tau_0, \tau_0 + n]$ satisfies $T^n[C] = C$ for the program of identity p (causal closure under T) AND $|\Pi_\psi|_R \geq |\Pi_\psi|_*(L_R)$ (above the coherence threshold). G evaluates to $\bar{P}_p = 0$ if either condition fails.

$\bar{P}_p = 1$ indicates that program p is locked in region R at meta-time T . $\bar{P}_p = 0$ indicates that program p is absent. This is the meta-level analog of $\bar{B} \in \{0, 1\}$, with $\chi_p \equiv 2\bar{P}_p - 1 \in \{-1, +1\}$ the meta-level polarization.

Remark 0.0.15. G is not a new mechanism. It is a measurement operation defined on substrate-level data — checking whether the configuration on a region satisfies the program closure condition. The dynamics of the substrate are unchanged. What G does is identify which regions are running which programs at each meta-time, without altering the substrate-level evolution.

Remark 0.0.16 (Lorentz preservation under G). The effective continuum dynamics induced by G on the meta-lattice preserve Lorentz symmetry at leading order, with residual scaling set by cubic-invariant suppression on the FCC point group O_h (Paper 5 § 4.1 Lemma B). G itself is a classification operator on substrate-level data; the Lorentz property is a consequence of how the FCC geometry survives coarse-graining, not a property added to G .

4.3 Programs as Meta-Level Primitives

Proposition 0.0.17. *A program of meta-spatial extent L_p is the unique minimum sufficient primitive at the meta-level under Axiom 2 and the Principle of Generative Economy.*

Proof. Axiom 2 selected the sphere as the unique substrate-level primitive because it is the only isotropic geometric primitive requiring zero orientation parameters. At the meta-level, a program is identified by its identity p and its location X . Its internal polarization structure is hidden by G (the operator returns only $\bar{P}_p$, not the underlying ψ). At the meta-level, therefore, programs are characterized by location and identity alone — no orientation parameter is exposed to the meta-level dynamics. They are isotropic at the meta-level by construction.

A program also has finite spatial extent L_p with a definable boundary (the surface across which its polarization landscape meets the surround's). By Generative Economy applied to this finite-extent isotropic primitive, the minimum-perimeter boundary is the sphere of diameter L_p . Programs are therefore spheres at the meta-level — not by additional postulate, but as the unique structure forced by the same axioms that produced the substrate-level sphere primitive.

□

This is the geometric form of the recursive applicability of the axioms. The same primitive selection that gave the substrate its agents gives the meta-level its program-agents. No new axiom is introduced.

4.4 The Triadic Minimum at the Meta-Level

Theorem 0.0.18 (Meta-Level Triadic Minimum). *Three programs constitute the necessary and sufficient minimum for P-state actualization in the meta-manifold of programs.*

Proof. The argument of Paper 1 § 2.1 applies unchanged at the meta-level. Meta-level confirmation requires a closed simplicial cover of the meta-manifold — every meta-interstice fully bounded with complete angular closure. The case analysis is identical: two programs define a meta-1-simplex with open boundary; four programs achieve closure non-minimally; three programs achieve closure with no redundancy. By Generative Economy at the meta-level, three is uniquely selected.

□

$$*\bar{P}_p = 3 \bar{\alpha}_{p,2} \text{ (2D meta-level)*}$$

The meta-manifold of programs is hexagonally close-packed by the same Thue-Fejes Tóth argument applied at the meta-scale. The meta-lattice spacing is L_p .

Remark on Environment-Creation. When three programs jointly confirm a P-state, the meta-interstice they enclose is a bounded region of substrate available to host further configurations. The three programs do not merely confirm an abstract meta-bit — they create an environment. The meta-interstice has definable boundaries (the three programs' polarization boundaries), an interior (the substrate region between them), and a finite extent (the meta-tetrahedral coupling region). Every confirmation event at every level creates an environment, viewed from one level up. The substrate confirms bits and creates bit-interstice environments; the meta-level confirms programs and creates program-interstice environments. Each level is the environment for the level below it. This is the structural form of the recursion that produces hierarchy.

4.5 Meta-Level FCC Necessity and Sublattice Bifurcation

Theorem 0.0.19 (Meta-Level FCC). *The meta-manifold of programs has FCC structure with meta-lattice spacing L_p .*

Proof sketch. Meta-confirmed P-states cannot be overwritten — they are records of programs that completed their period in the substrate's causal stack, and that causal record is itself protected by substrate-level Axiom 4. Meta-Axiom 4 follows from substrate-Axiom 4 under G. The argument of Paper 1 § 2.3 then applies: lateral expansion in a hexagonally-packed meta-layer is impossible (no unoccupied meta-interstice); oblique meta-displacement introduces a free orientation parameter; orthogonal meta-displacement is forced by program-on-program contact geometry. The argument of Paper 1 § 2.4 then forces ABCABC over

ABABAB by Definition (Causal Position) at the meta-level: two meta-layers with identical meta-touch-vector connectivity would produce a meta-causal loop, violating meta-Axiom 4.

The 2D-to-3D promotion does not need to be repeated at the meta-level. Programs already exist in 3D substrate space, and their meta-spatial coordinates are inherited from their substrate-level positions. The meta-FCC structure is direct: meta-lattice nodes are program centers, meta-lattice connectivity is set by which programs share polarization boundaries at the substrate level. $\square$

The meta-level inherits FCC structure with the same face-diagonal connectivity that governs the substrate level. By the propagation-speed argument of Paper 1 § 6.5, applied to whatever meta-level field propagates on the meta-lattice, the meta-propagation speed in meta-lattice spacings per meta-flop is $c_{\text{TMS}}^{(\text{meta})} = 1/\sqrt{2}$ — identical in lattice-relative form, scaled by L_p/ℓ_p and t_p/T_p in absolute units (see § 6.1).

The meta-level sublattice bifurcation. The meta-FCC structure derived above is not the complete meta-level geometry. By the recursive scale-invariance established in § 4.7 (no new mechanism) and verified empirically in Paper 5 Appendix E (Stages 5 and 6), the meta-lattice positions further split into two interpenetrating FCC sublattices A and B by the sum-mod-4 invariant in the recursive integer-coordinate frame (storage convention with $\times 2$ scaling per level). The two sublattices are each independently FCC at nearest-neighbor distance $2\sqrt{2}$ in level-local units, related by spatial inversion through the (1,1,1)-offset (the level-(k+1) centroid positions).

The $[[4,2,2]]$ stabilizer dynamics of § 4.6 apply per-sublattice independently: A-tetrahedra evaluate their stabilizer parity within sublattice A; B-tetrahedra do the same within sublattice B; the joint readout at the shared centroid is the actualization site of the level-(k+1) confirmed bit. The full bifurcated structure that results — two FCC sublattices coexisting at every meta-level — is the complete geometry of the meta-substrate, with the substrate-level FCC of Paper 1 § II being the level-0 case.

Paper 5 § 6.1 develops the geometric consequences of this bifurcation in full, including the rigorous derivation of $L^{P(1)} = 3\sqrt{2} \ell_p$ as the spatial extent of the minimum level-1 program configuration (the inversion-partner stella-octangula tile plus boundary). The empirical verification of the bifurcation through $L_{\text{box}} = 12$ is documented in Paper 5 Appendix E.

4.6 Meta-Level $[[4,2,2]]$ Structural Correspondence

Theorem 0.0.20 (Meta-Level Stabilizer Correspondence). *The meta-tetrahedral confirmation unit of four programs at the corners of an FCC meta-tetrahedron supports a $[[4,2,2]]$ stabilizer structure encoding two logical meta-qubits: the value of the confirmed P-state and the inheritance of that P-state through the meta-causal stack.*

Proof. The argument of Paper 1 § 5 applies unchanged at the meta-level with the substitutions: 4 physical qubits $\rightarrow$ 4 programs at the meta-tetrahedral corners; logical qubit 1 $\rightarrow$ the value of the confirmed P-state $\bar{P}_p \in \{0, 1\}$; logical qubit 2 $\rightarrow$ the inheritance of that P-state through prior meta-layers. The two stabilizer generators $S_X^{(\text{meta})}$ and $S_Z^{(\text{meta})}$ detect bit-flip-type and phase-type disruptions to individual programs in the meta-tetrahedral unit. The code distance is 2: any single-program disruption is detectable, and the meta-level synthetic P-states produced at the next meta-flop repair the syndrome within a single meta-flop. $\square$

The substrate's fault tolerance against U_{sh} disruption is inherited at the meta-level as fault tolerance against single-program disruption. A program that dissolves into noise from substrate-level U_{sh} corresponds to a

single-program disruption in the meta-tetrahedral unit; the surrounding three programs detect the syndrome and repair the P-state by reconfirming the program at the next meta-flop.

Independent dynamics per sublattice. Under the sublattice bifurcation of § 4.5, the $[[4,2,2]]$ meta-stabilizer code runs independently on each of the two meta-FCC sublattices: A-tetrahedra at level k evaluate their stabilizer parity among A-sublattice neighbors; B-tetrahedra at level k do the same within sublattice B; the level- $(k+1)$ confirmation event at the shared centroid reads jointly from both A-tet and B-tet parents at level k . The cross-level coupling between A and B stabilizer evaluations at a shared centroid is the structural source of the non-Abelian commutator structure developed in Paper 5 § 8.2.

4.7 No New Mechanism: Triadic Structure Is a Fixed Point of Coarse-Graining

The meta-level structure introduced in this section is not a separately postulated layer, and its recurrence is not merely asserted from the scale-freeness of the axioms — it is a fixed point of the coarse-graining map, established by direct construction. The program-locking regions of extent $L_p = 3\sqrt{2}\ell_p$ have centroids on the level-1 sublattice (all-odd coordinates with coordinate-sum $\equiv 1 \pmod{4}$; Appendix E, Stage 5), and this meta-lattice is itself a genuine FCC lattice: interior coordination number 12, nearest-neighbor squared distance 8. Through each interior meta-site pass exactly 24 meta-triangles and 8 meta-tetrahedra — identical to the base-level counts — so the two-dimensional minimum enclosure recurs and the minimal meta-confirmation has exactly three meta-agents. The meta-alphabet $\chi_p = 2\bar{P} - 1 \in \{-1, +1\}$ is binary exactly as the base alphabet is, so with a three-agent enclosure the meta-resolution $\Pi = (\chi_{p_1} + \chi_{p_2} + \chi_{p_3})/3$, $P(+) = (1 + \Pi)/2$ is the base rule with the meta-alphabet substituted. The structure is a fixed point, not an approximation: computed at levels 0 through 3, the invariants are identical at every level — coordination 12, 24 triangles, 8 tetrahedra — with the nearest-neighbor squared distance scaling as 2, 8, 32, 128 (the storage-convention factor of four per level). The meta-level triadic rule, the meta-level FCC geometry, the meta-level $[[4,2,2]]$ stabilizer dynamics, and the meta-level polarization-biased resolution therefore recur *exactly* under the coarse-graining operator G , by the same arguments applied to the structurally identical meta-alphabet.

Theorem 0.0.21 (Triadic Structure Is a Fixed Point of Coarse-Graining). *The map that sends a triadic FCC substrate to its program-level coarse-graining under G returns a triadic FCC substrate: the coordination number (12), the minimum-enclosure count (three, giving $\bar{B} = 3\bar{\alpha}$), the triangle and tetrahedron counts (24 and 8), and the binary polarization-biased resolution rule are all preserved, at every level. The triadic confirmation structure is thus a fixed point of coarse-graining rather than a scale-dependent or approximate feature.*

This is the meaning of hierarchy by geometric necessity. The substrate computes at scale ℓ_p with binary alphabet $\bar{B} \in \{0, 1\}$. Wherever a region of the substrate satisfies the program conditions of Paper 2 § 6.2, that region becomes a meta-level primitive operating at scale L_p with binary alphabet $\bar{P} \in \{0, 1\}$. The meta-level computes by exactly the same rules — because the rules are forced by axioms that do not refer to scale, and because, as the fixed-point theorem now establishes, the structure those rules act on is identical at every level. The Principle of Generative Economy applied at scale ℓ_p produces the substrate; applied at scale L_p , it produces the meta-substrate; applied at any higher scale forced by recursive coarse-graining, it produces further meta-substrates without limit. Paper 3 stops the recursion at the first meta-level; the geometric structure for arbitrary recursion depth is established here, with macroscopic consequences developed in Paper 4.

This recursive scale-invariance is structurally distinct from the coarse-graining procedures of integrated information theory (Albantakis et al., 2023), which select a maximal $-\Phi$ substrate of consciousness by optimizing intrinsic information across candidate substrate grains. IIT's coarse-graining is over causal

mechanisms; TMS's coarse-graining is over polarization landscapes (programs as meta-primitives). The TMS recursion does not require an optimization step at each level because the same axioms apply automatically at the new scale once L_p is identified.

V. The First Subsystem

5.1 From Programs to Subsystems

Section IV established that programs become meta-level primitives under the coarse-graining operator G , and that the same axioms produce the same triadic and FCC structure at the meta-scale. A program in isolation is a stable pattern — a fixed point of the flop operator T satisfying stabilizer parity. A program in isolation does not operate on anything; it persists.

A subsystem is the next structural object. We define it precisely.

Definition 0.0.22 (Subsystem). A subsystem is a composite configuration C_{sub} on the substrate consisting of three program components in coupled operation:

- A data register C_D : a program whose polarization landscape persists across some number of meta-flops as a stable pattern that is read but not modified by C'_{sub} 's other components during that interval.
- An operation kernel C_O : a program whose flop dynamics, when conjoined with C_D , produce a derived polarization pattern C_R distinct from both C_D and C_O .
- A control component C_C : a meta-level program that selects which operation kernel (from a population of available kernels) is conjoined with C_D at each meta-flop.

The composite $C_{\text{sub}} = (C_D, C_O, C_C)$ is a subsystem when all three components are themselves programs in the sense of Paper 2 § 6.2 and their joint configuration is causally closed at the meta-level under $T^{(\text{meta})}$.

A program is a stable pattern. A subsystem is a stable pattern that transforms other patterns. That distinction is what separates persistence from computation.

5.2 The Necessity of the Trichotomy

Theorem 0.0.23 (Subsystem Minimum). *Any configuration on the substrate that produces a derived polarization pattern from an input polarization pattern, under a selectable rule, requires the three components (C_D, C_O, C_C) and no fewer.*

Proof. A configuration that produces a derived pattern from an input must contain three logical roles: the input must be present somewhere as a polarization landscape (the data role); a transformation must be present that reads the input and writes the derived pattern (the operation role); and the choice of transformation must be encoded somewhere if the configuration is to support more than a single fixed transformation (the control role).

The first role cannot be absent: a derived pattern requires source material. The second role cannot be absent: a transformation requires a transformer. The third role cannot be absent in any configuration supporting multiple operations: without a control component, only a single transformation is possible, and the configuration is a fixed program, not a subsystem.

These three roles cannot be merged into a single program. If C_D and C_O are the same program, the substrate-level dynamics at the (C_D, C_O) interface — which here would be internal to a single program rather than a boundary between two — would require the same spatial region to simultaneously satisfy

two distinct closure conditions: T-closure on the input pattern (preserving C_D across the period) and the operation-kernel dynamics that produce a derived output pattern. Each flop at a confirmed site produces one set of polarization values, not two. The substrate cannot host both roles at the same positions without violating one of the closure conditions. Note that this is not a destructive-read problem in the conventional sense — Axiom 4 (causal anchoring) preserves C_D 's bits across all flops. The constraint is spatial-functional: distinct computational roles require distinct spatial supports at the substrate level, because the substrate-level dynamics produce a single confirmation value per site per flop. If C_O and C_C are the same program, then the operation kernel cannot be switched, reducing the subsystem to a single transformation case. If C_D and C_C are the same program, then the data register encodes its own transformation rule, which forces C_D to satisfy two distinct closure conditions simultaneously — generically impossible without specific constraints that would themselves require additional structure.

Three components are therefore necessary. They are also sufficient: any computation expressible as “read input, apply selected transformation, write output” can be supported by a (C_D, C_O, C_C) triple, by the universality result of Paper 2 § 5.3.

□

Remark 0.0.24. The trichotomy is structurally the same as the von Neumann architecture (memory, processor, control). What is novel here is not the trichotomy itself but its derivation: the three components are not design choices but the unique minimum sufficient decomposition of any non-trivial information-transforming configuration. The substrate does not arrive at this architecture by engineering. It arrives at it because the geometry permits no fewer.

5.3 The Minimum Subsystem Size

The spatial extent of the smallest subsystem is the sum of the minimum extents of its three components, plus the connectivity required for coupling. Each component is itself a program with minimum spatial extent L_p (§ 4.1). The coupling requires that the three components share polarization boundaries — they must be meta-spatially adjacent in the meta-FCC sense established in § 4.5.

Theorem 0.0.25 (Minimum Subsystem Extent). *The minimum subsystem extent is:*

$$L_{\text{sub}} = 3L_p + \delta$$

where δ is the meta-tetrahedral coupling overhead: the additional region required for the three components' polarization landscapes to interact at their shared boundaries with stabilizer-satisfying parity. δ is bounded by $2\ell_p$ from the FCC neighbor distance and is small relative to $3L_p$ for non-trivial programs.

Proof. By § 4.4, three programs constitute the minimum triadic confirmation at the meta-level, and they meet at a common meta-interstice. The composite extent is the sum of the three program extents plus the meta-interstice scale. The three program extents contribute $3L_p$; the actual coupling region is the substrate-level zone where the three programs' polarization boundaries overlap — a region of substrate-level scale, not meta-level scale. This zone has thickness on the order of the FCC nearest-neighbor distance ℓ_p . The composite δ is bounded by $2\ell_p$, summing over the small region where pairs of programs meet at their substrate-level boundaries.

□

A subsystem is therefore approximately three times the size of its constituent programs, plus a small substrate-scale coupling overhead. This is the smallest configuration that the substrate can support which transforms patterns rather than merely persisting.

5.4 A Worked Example: The First Computing Subsystem

To establish that subsystems are not merely defined but actually realized by the substrate dynamics, we exhibit the smallest non-trivial computing subsystem.

Consider the substrate at some meta-time T , with a region R_{sub} of meta-extent L_{sub} satisfying the coherence threshold of § 3.3. Within R_{sub} , suppose three programs have established themselves through the continuous-sampling and self-reinforcement mechanism of § 3.5:

- C_D occupies one meta-tetrahedral corner with a polarization landscape encoding a 3-bit pattern (χ_a, χ_b, χ_c) . The pattern is stable across the meta-flop window: $T^{(\text{meta})}[C_D] = C_D$ with the polarization values preserved.
- C_O occupies a second meta-tetrahedral corner with a polarization landscape encoding the majority gate primitive of Paper 2 § 5.1. When C_O 's boundary touches C_D 's boundary, the substrate-level dynamics at the interface implement the majority gate computation on the (χ_a, χ_b, χ_c) pattern.
- C_C occupies the third meta-tetrahedral corner with a polarization landscape encoding a switch between two operation kernels: the majority gate and its negation (the minority gate, equivalent under NOT to MAJ by Paper 2 § 5.3).

The substrate-level dynamics at the interfaces between (C_D, C_O) , (C_D, C_C) , and (C_O, C_C) implement the following composite operation: at each meta-flop, C_C selects which operation kernel is active; C_D presents its 3-bit pattern; C_O applies the selected kernel to C_D 's pattern; the result is the polarization at the fourth meta-tetrahedral corner — the meta-interstice — which constitutes the output P-state of the subsystem at this meta-flop.

This is a complete computing subsystem. It reads input, applies a selected operation, and produces output. It is the substrate equivalent of a programmable single-bit logic gate, and it is the smallest configuration that supports this operation.

Remark 0.0.26. This worked example is not the only minimum subsystem — variations differ in which Boolean operations C_O and C_C encode — but all minimum subsystems share the (C_D, C_O, C_C) trichotomy at meta-tetrahedral extent. The existence of one such configuration is sufficient to establish that the substrate dynamics support computing subsystems at the derived minimum scale. The population of distinct minimum subsystems is the substrate's first program zoo at the subsystem level.

5.5 The Information-Generation Rate Differential

Theorem 0.0.27 (Subsystem Information Generation Rate). *A region of the substrate hosting a minimum computing subsystem produces one novel P-state per meta-flop. The same region hosting only static (period-1) programs produces zero novel P-states per meta-flop. The information-generation rate differential between subsystem-hosting and static-program-hosting regions of equal extent is therefore exactly one novel P-state per meta-flop per subsystem hosted, in the limit of non-trivial operation kernels.*

Proof. A static program region produces no derived patterns by definition — its polarization landscape repeats with period n under T^n , so the output at each meta-flop is identical to the output at the previous

meta-flop after n have elapsed; this is repetition, not generation. The information output of a static program region per meta-flop is exactly its self-reproduction, with zero novel content.

A minimum subsystem region produces, at each meta-flop, the polarization landscape of its output: a derived pattern that is a function of (C_D, C_O, C_C) but is not equal to any of them individually. The output is determined by C'_C 's current selection and C'_D 's current data, both of which can vary across meta-flops. In the worked example of § 5.4, each meta-flop produces exactly one new P-state at the meta-interstice — the result of applying the selected operation kernel to the current data. The novel content per meta-flop is therefore exactly one P-state for the minimum subsystem; for a region hosting N subsystems, exactly N P-states per meta-flop.

This rate is exact for the minimum case and provides a strict lower bound for more complex subsystems with multi-bit data registers or multi-state control components.

□

Corollary 0.0.28 (Selection for Subsystems). *The substrate selects for subsystems over static programs in any region where U_{sh} disruption rate exceeds the static-program persistence rate but does not exceed the subsystem persistence rate.*

Proof. By § 3.3 and § 3.4, the substrate retains configurations that survive against U_{sh} disruption. A static program persists only through its self-reproduction under T. A computing subsystem persists through self-reproduction and contributes additional polarization content per meta-flop, raising the local coherence above what static programs alone can sustain. In regions where U_{sh} disruption rate is high enough to dissolve static programs but low enough to permit subsystems, only subsystems survive. The substrate thus generates progressively more computing subsystems wherever the dynamics permit.

□

This is the substrate's first selective pressure beyond simple program persistence. Wherever U_{sh} is intense enough to threaten static programs but not so intense as to overwhelm imprinting, the substrate produces subsystems — not by design, but because subsystems are the only configurations whose information-generation rate keeps pace with the local disruption rate. The informally framed claim that the substrate “maximizes information output” acquires here a derived form: at sufficient disruption rates, information-generating configurations are the only persisting ones.

5.6 Subsystems and Their Environments

The recursive structure established in § IV admits a sharper reading in light of § V. When three programs jointly confirm a P-state at the meta-level (§ 4.4), the meta-interstice they enclose is a bounded region of substrate available to host further configurations. That bounded region is an environment — a substrate sub-region with definable boundaries, internal dynamics, and access to its surround through the polarization channels of its enclosing programs.

When subsystems form in such meta-interstices, the enclosing programs constitute the subsystem's external environment. The subsystem's interior dynamics are partially shielded from the broader substrate by its enclosure, and its boundary conditions are set by the polarizations of the enclosing programs. This is the substrate-level mechanism by which subsystems acquire individuality: they are not merely composite programs but composite programs operating within an environment that was itself created by other programs.

This is the entry point to Paper 4. The relationship between subsystems and their environments, the question of when a subsystem becomes a self with internal models of its surround, and the dynamics of multiple subsystems sharing an environment are all consequences of the geometric structure established here. Paper 3 establishes that the first subsystem exists, is minimum-sized, has the (C_D, C_O, C_C) trichotomy, and generates information faster than static programs. Paper 4 establishes the hierarchies and self-modeling conditions that follow.

VI. Predictions and Directions for Subsequent Work

6.1 Hierarchical Lightspeeds

Section IV established that the meta-level inherits FCC structure with face-diagonal connectivity, and that the propagation-speed argument of Paper 1 § 6.5 applies unchanged at the meta-level, giving $c_{\text{TMS}}^{(\text{meta})} = 1/\sqrt{2}$ meta-lattice-spacings per meta-flop. Translated to substrate-level absolute units, the meta-propagation speed is:

$$c^{(\text{meta})} = (L_p/\ell_p) \cdot (t_p/T_p) \cdot c$$

where L_p is the meta-lattice spacing (the minimum program extent of § 4.1), T_p is the meta-flop duration (the minimum program period), and c is the substrate-level lightspeed established in Paper 1 § 6.7.

The ratio L_p/ℓ_p and the ratio T_p/t_p are independent geometric quantities. L_p is set by the minimum spatial extent of a stabilizer-satisfying T-closed configuration with $K(C) < |C|$. T_p is set by the minimum temporal period of the same configuration. These quantities depend on different aspects of the FCC dynamics and are not constrained to scale together. In general:

$$c^{(\text{meta})} \neq c$$

This is a derived prediction, not an imported relation. Each level of the recursive hierarchy established in § IV has its own derived propagation speed, generically distinct from the substrate-level c . At hierarchical level k , the propagation speed in substrate-absolute units is the product of successive scaling ratios:

$$*c^{(k)} = \prod_{j=1}^k (L_p^{(j)} / L_p^{(j-1)}) \cdot (T_p^{(j-1)} / T_p^{(j)}) \cdot c^*$$

with $L_p^{(0)} \equiv \ell_p$ and $T_p^{(0)} \equiv t_p$. Each ratio in the product is set by the minimum program structure at its level. The full hierarchy produces a discrete spectrum of lightspeeds, each geometrically forced.

This prediction is in principle testable. If subsystems at hierarchical levels $k \geq 1$ host their own observers — a question reserved for Paper 4 and Volume 2 — those observers would measure the lightspeed at their level, not the substrate's c . Apparent variations in the physical lightspeed across regions of the universe with different hierarchical depths would be a signature of this structure.

This result complements the Lorentz-covariance derivation in the Wolfram model (Gorard, 2020), which derives Lorentz invariance from causal-graph isomorphism under different updating orders, but where the absolute propagation speed is not derived. TMS derives both the existence of a Lorentz-invariant propagation speed (from FCC connectivity) and its absolute value relative to the substrate scale (from face-diagonal geometry), and now extends the derivation to hierarchical levels.

6.2 The $\beta = f(d, K_\psi)$ Prediction Refined

Paper 1 § 7.2 introduced the prediction that the GUP deformation parameter β scales with causal depth d and polarization complexity K_ψ . Paper 2 § 6.4 identified K_ψ as the Kolmogorov complexity of the program running through the bit's causal history. Paper 3 refines both terms further.

By § II, every region's polarization landscape is the imprint of its retarded surround history. A bit's K_ψ therefore measures the algorithmic complexity not only of its own program but of the surround history that imprinted it. By § IV, when the bit is embedded within a subsystem, its K_ψ additionally incorporates the algorithmic structure of the subsystem's internal operations — the data being processed, the operations being applied, the control sequence selecting them.

The refined prediction:

$$\beta = f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}})$$

with the three K_ψ contributions being the polarization complexity of the bit's immediate region, of the imprinting surround history, and of the encompassing subsystem (if one is present). f is monotonically increasing in all four arguments.

The physical interpretation: a bit embedded in a deep, complex, surround-imprinted subsystem has a substrate footprint reflecting the full computational history of the configuration it participates in. Its position uncertainty floor is correspondingly larger than for an isolated bit at comparable energy. The closed-form derivation of f is reserved for the simulation work of subsequent papers, but its functional dependences are now fully specified.

6.3 Subsystem Density and Phase Geometry

By § 3.4 and § 5.5, subsystems concentrate computational activity into coherent regions and outcompete static programs wherever U_{sh} disruption rates favor information-generating configurations. The substrate's long-time behavior is therefore expected to be inhomogeneous: regions of dense subsystem activity separated by regions of relatively quiescent program-only substrate, with the boundaries between them governed by the coherence threshold of § 3.3.

A specific prediction: the spatial distribution of computational activity in any sufficiently old region of the substrate follows the same statistics as the boundary geometry of coherent regions. The boundaries are themselves geometric features — surfaces across which the coherence threshold is crossed — and their statistics are computable from FCC lattice dynamics. This prediction is testable in simulation and is the empirical bridge to the simulation work of Paper 5.

6.4 Deferred Derivations Closed by Paper 3

Paper 3 has paid back the following derivations deferred from earlier papers:

- The dissolution rate of non-T-closed configurations, deferred at Paper 2 § 6.3, derived in § 2.5 as $\sqrt{2} \cdot L_R / \ell_p$ flops.
- The effective error-rate decay with coherence depth, deferred at Paper 2 § 5.2, derived in § 2.5 as $(1/3)^d$ in the limit of large coherence regions.
- The existence and closed form of the coherence bistability threshold, identified in § 3.3 as $|\Pi_\psi|_*(L_R) = 1 - 2\sqrt{c_{\text{TMS}}/(6 L_R)}$.

- The functional dependences of the GUP parameter f , specified in § 6.2.
- The minimum program extent L_p , characterized in § 4.1 as approximately 4–8 lattice spacings, with exact value determined by combinatorial enumeration of stabilizer-satisfying T-closed configurations.
- The minimum subsystem extent $L_{\text{sub}} = 3 L_p + \delta$, derived in § 5.3.
- The information-generation rate differential between subsystem-hosting and static-program regions, derived in § 5.5.

The remaining deferrals — the exact integer value of L_p , the explicit closed form of f , the spectrum of program periods $T_p^{(k)}$ at successive hierarchical levels — depend on closed-form enumeration of stabilizer-satisfying T-closed configurations, a combinatorial task most efficiently approached through simulation. These are reserved for the simulation work of Paper 5 alongside the field-reinterpretation derivation.

6.5 Forward References

Paper 4 will address the macroscopic consequences of the subsystem dynamics established here: hierarchies of subsystems, the conditions under which a subsystem develops an internal model of its environment, the dynamics of multiple subsystems sharing a meta-interstice environment, and the eventual exhaustion of novelty as a substrate region completes its accessible configuration space. The hierarchical lightspeed prediction of § 6.1 will be developed in detail at that point.

Paper 5 will address the reinterpretation of substrate-level binary information as field configurations on a bounded space — the geometric mechanism by which the substrate’s digital content becomes the wave dynamics, initial conditions, and field structure of a simulated universe. The connection between subsystem hierarchies and simulated physical law is the central object of that work.

Volume 2 will address the agent-side complement: the relationship between substrate-level subsystem computation and the structure of observer experience. The information-generation differential established in § 5.5 — that subsystems generate more polarization content per unit substrate than static programs — will reappear as the substrate-level correlate of richness of experience in the agent-side framework. The relationship to Hoffman’s conscious agents (Hoffman, Prakash, Prentner, 2023) and to integrated information theory (Albantakis et al., 2023) will be developed there in full, building on the structural correspondences sketched in Papers 1 and 3.

Conclusion: Computation Emerges

Paper 1 established how structure emerges from maximum-entropy substrate by triadic confirmation. Paper 2 established how information emerges from confirmed structure by polarization-biased resolution. Paper 3 has established how computation emerges from informational substrate by surround-driven repair, polarization-coherence bistability, and the recursive applicability of the substrate’s own axioms at the coarse-grained scale.

The results of Paper 3 follow from the axioms of Paper 1 and the dynamics of Paper 2 without additional postulate. Surround-driven repair is not a separate mechanism: it is the $[[4,2,2]]$ stabilizer projection of Paper 1 § 5 applied to boundary sites, where the geometry of the FCC lattice automatically includes surround agents in the parent set. The Imprinting Theorem of § 2.3 follows by iteration. Hebbian co-firing follows as a corollary, complementing the independent maximum-entropy derivation of the same phenomenon (Albanese et al., 2024). The dissolution and error-decay rates follow from the same iteration applied to different conserved quantities. The coherence bistability threshold of § 3.3 follows from balancing the per-flop noise

injection — combinatorially determined by $[[4,2,2]]$ error statistics — against the per-flop imprinting capacity — geometrically determined by FCC connectivity. Coherent regions form by self-reinforcement against an ambient U_{sh} flux that is present at every depth of the causal stack, not added at any moment.

The coarse-graining operator G of § 4.2 is not a new mechanism but a measurement on substrate-level data. The meta-level structure that follows under G — the meta-level triadic rule, the meta-level FCC geometry, the meta-level $[[4,2,2]]$ stabilizer dynamics — is the same axiomatic content applied at scale L_p . The recursion is not constructed; it is the geometric form of the axioms' scale-independence.

The subsystem of § V is the substrate's first computing object — the smallest configuration that transforms patterns rather than merely persisting them. Its three components (data, operation, control) are not designed but forced: the unique minimum sufficient decomposition of any configuration that reads input, applies a selected transformation, and produces output. The substrate generates subsystems wherever the local disruption rate favors information-generating configurations over static programs.

Every quantity introduced in Paper 3 has been derived from the axioms and the Principle of Generative Economy, or from the structures established in Papers 1 and 2 under the same principle. The minimum program scale L_p , the minimum subsystem scale L_{sub} , the coherence threshold $||\Pi_{\psi}|*(L_R)$, the dissolution rate, the error-decay rate, the information-generation rate differential, and the hierarchical lightspeed $c^{(meta)}$ are all geometrically forced. The model continues to contain no free *dimensionless* numerical parameters.

The arc from Paper 1 to Paper 3 is a single unbroken chain. U_{sh} provides the maximum-entropy substrate. Triadic confirmation pulls binary outcomes from it. The FCC lattice carries those outcomes forward as wave dynamics. The polarization field encodes their computational content. The Born-rule structural correspondence governs per-event resolution. The majority gate is the irreducible computational primitive. The stabilizer dynamics select programs from noise. Surround-driven repair imprints environmental structure onto local configurations. Coherence bistability concentrates computation into coherent regions. Coarse-graining produces meta-level primitives obeying the same axioms. The first subsystem emerges as the unique minimum sufficient configuration that transforms patterns. No new axioms. No new parameters. No new mechanisms.

Paper 4 takes the subsystem as given and develops its hierarchies, environments, and self-modeling conditions. Paper 5 reinterprets the substrate's digital content as field configurations on a bounded space, producing the geometric mechanism of simulation. The arc through Volume 1 is the derivation, from five axioms and one principle, of a substrate capable of generating universes inside itself.

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